Lecture 0 – Course Overview

COSE215: Theory of Computation

Jihyeok Park

PLRG

2026 Spring

Course Information

- **Instructor:** Jihyeok Park (박지혁)
  - **Position:** Assistant Professor in CS, Korea University
  - **Expertise:** Programming Languages, Software Analysis
  - **Office hours:** 14:00–16:00, Tuesdays (appointment by e-mail)
  - **Office:** 507, Jung Woonoh IT & General Education Center
  - **Email:** jihyeok_park@korea.ac.kr
- **Class:** COSE215 - 02 (English)
- **Homepage:** https://plrg.korea.ac.kr/courses/cose215/
- **LMS:** https://lms.korea.ac.kr/
  - Please use the **Board** > **Q&A** section for questions.
- **Teaching Assistant:** cose215@googlegroups.com
  - Hyunjoon Kim (김현준)
  - Minseok Choe (최민석)
  - Sungmin Park (박성민)

Grading

- **4 Homework Assignments: 20%**
  - Programming assignments in Scala (submission in **LMS**)
  - Please check the course website for the detailed **policy on academic integrity**. Some highlights are:
    - Do not share your code with others / Do not see others' code.
    - It's your code only if you can explain all the details of the code.
  - If violated, you get a **zero** for the assignment or an **F for the course**.
- **Midterm exam: 30%**
  - April 23 (Wed.) 13:30 – 14:45 (in class, 75 min.)
- **Final exam: 30%**
  - June 23 (Mon.) 13:30 – 14:45 (in class, 75 min.)
- **Attendance: 10%**
  - Please use **LMS** to attend the class with the code provided.
  - The first two attendance checks (Mar. 4 and 11) are just for practice.

Schedule

Weak	Contents	Weak	Contents
1	Basic Concepts	9	Pushdown Automata
2	Deterministic Finite Automata (DFA)	10	Deterministic Pushdown Automata
3	Nondeterministic Finite Automata (NFA)	11	Properties of Context-Free Languages
4	Regular Expressions and Languages	12	Turing Machines (TMs)
5	Properties of Regular Languages	13	Extensions of Turing Machines
6	Context-Free Grammars and Languages	14	Undecidability
7	Parse Trees and Ambiguity	15	P, NP, and NP-Completeness
8	**Midterm Exam (Apr. 22 - Wed.)**	16	**Final Exam (Jun. 22 - Mon.)**

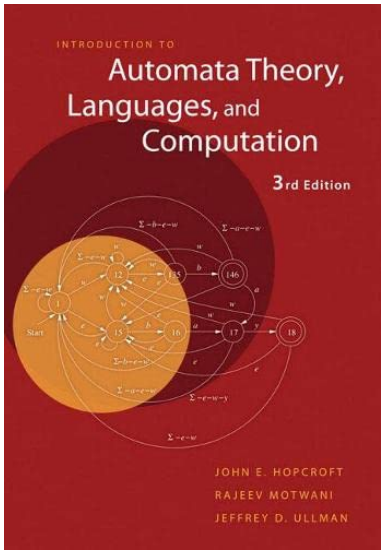
- There will be no offline lectures on following days.
  - May 25 (Mon.) – National Holiday (부처님 오신 날)
  - June 3 (Wed.) – Election Day (지방선거)
- Instead, recorded lecture videos will be uploaded to **LMS**.
- You don't need to check the attendance on these days.

Course Materials

- **Self-contained lecture notes.**

https://plrg.korea.ac.kr/courses/cose215/

- Reference:



John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman. Introduction to automata theory, languages, and computation. Third edition.

Goal of This Course

- What is the *mathematical model* of computers?

Turing Machine!

Let's learn **Turing Machine**

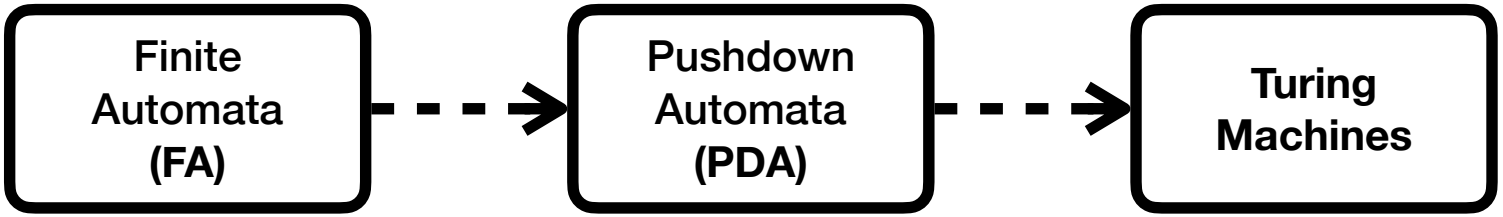
- Is it possible to solve *every problem* using computers?

No!

Let's learn **Undecidability** and **Intractability**

Roadmap: Towards Turing Machine

A Turing machine is a specific kind of **automaton**.



- **Part 1: Finite Automata (FA)**
  - Regular Expressions (REs)
  - Regular Languages (RLs)
  - Applications: text search, etc.
- **Part 2: Pushdown Automata (PDA)**
  - Context-Free Grammars (CFGs)
  - Context-Free Languages (CFLs)
  - Applications: programming languages, natural language processing, etc.
- **Part 3: Turing Machines (TMs)**
  - Lambda Calculus (LC)
  - Recursively Enumerable Languages (RELs)
  - Undecidability and Intractability

Roadmap: Towards Turing Machine

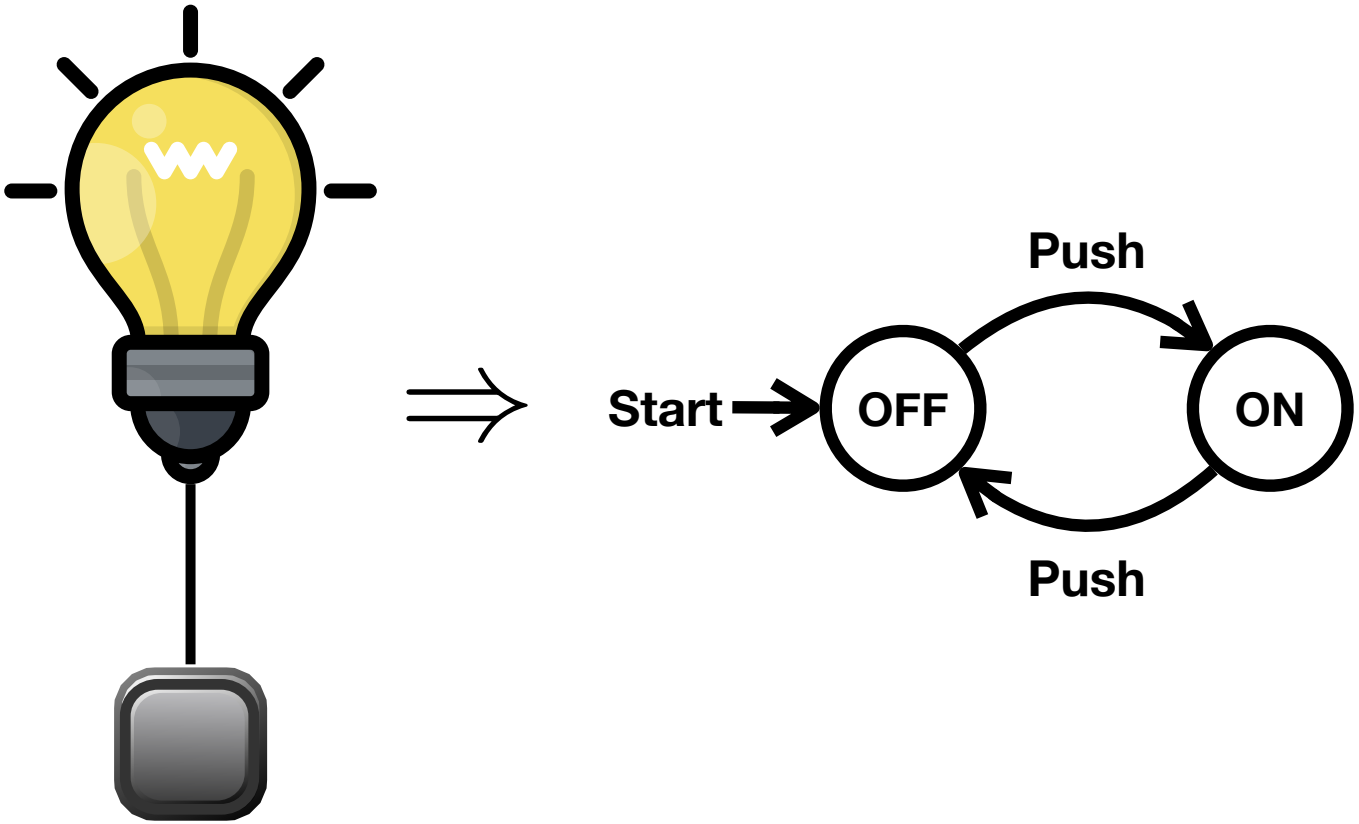
	Automata	Grammars	Languages
(Part 3) Turing Machines	(Lecture 23) ETM $\rightleftarrows$ (Lecture 21/22) TM $\rightleftarrows$	(Lecture 24) LC	(Lecture 21) REL $\cup$ DL $\supset$ (Lecture 25) **NP** $\overset{=}{\underset{\supset}{?}}$ **P** (Lecture 26)
(Part 2) Pushdown Automata	(Lecture 14/15) $\mathrm{PDA_{FS}} \rightleftarrows \mathrm{PDA_{ES}}$ (Lecture 16) $\cup$ $\mathrm{DPDA_{FS}} \supset \mathrm{DPDA_{ES}}$ $\cup$ (Lecture 17) $\not\supset$	(Lecture 11/12) CFG **Chomsky Normal Form** (Lecture 18)	(Lecture 11) CFL **Closure Properties** (Lecture 19) (Lecture 13) **Parse Trees & Ambiguity** **Pumping Lemma** (Lecture 20)
(Part 1) Finite Automata	(Lecture 4) NFA $\rightleftarrows$ (Lecture 3) DFA $\rightleftarrows$ (Lecture 5) ϵ-NFA (Lecture 7) **Equivalence & Minimization** (Lecture 10)	(Lecture 6) RE	(Lecture 3) RL **Closure Properties** (Lecture 8) **Pumping Lemma** (Lecture 9)
(Part 0) Basic Concepts	(Lecture 1) **Mathematical Preliminaries**	(Lecture 2) **Scala**	

Introduction of Automata

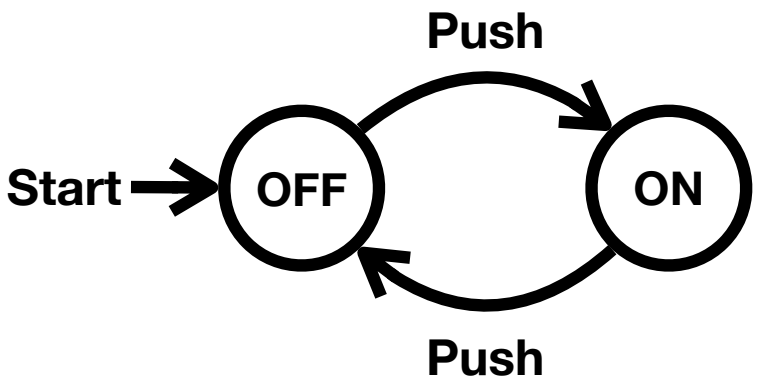
A Turing machine is a specific kind of **automaton**.

Then, what is an **automaton**? A **state transition system** that takes an **input** and changes its **state** based on the input.

For example,



Introduction of Automata



Theorem

The current state is OFF *if and only if the button is pushed even times.*

- Is it possible to prove it?

  Let's learn **mathematical background and notation.**

- Is it possible to implement the automaton?

  Let's learn **Scala** as an implementation language.

Next Lecture

- Mathematical Preliminaries

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